

Rami Suissa

Firmware Engineer

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SUMMARY

Firmware engineer with 11 years of industry experience in embedded software for ARM Cortex-M platforms, after service in an IDF technological unit. Main areas: C, FreeRTOS, low-level drivers, bootloaders and OTA update mechanisms. Experienced in development under medical device regulation (IEC 62304).

EXPERIENCE

Senior Firmware Engineer · Corvantis (medical devices), Petah Tikva 2020 - Present

- Develop firmware in C for the infusion pump product line, based on STM32 (Cortex-M4) and FreeRTOS.
- Responsible for the bootloader and firmware update mechanism, including dual-bank OTA update over BLE with signed image verification.
- Wrote and maintain drivers for I2C, SPI and UART peripherals: sensors, EEPROM, motor controller, display.
- Implement safety mechanisms according to IEC 62304 Class C requirements: watchdog handling, RAM tests, CRC checks on configuration and code.
- Write software design documents, unit tests (Ceedling) and code review records as part of the regulatory file.
- Support the verification team during V&V cycles and handle firmware defects found in testing.
- Participated in preparation for two FDA audits as the firmware representative.
- Promoted to senior in 2023; guide two junior engineers on the team.

Firmware Engineer · Ionwave Semiconductors, Herzliya 2015 - 2020

- Developed embedded firmware for evaluation and production test boards of power management ICs (Cortex-M0/M3, bare metal and FreeRTOS).
- Wrote a UART/I2C based command protocol used by the validation teams for automated measurements.
- Implemented a flash bootloader with fallback image for field updates of test equipment.
- Maintained board bring-up code for new silicon revisions, including clock tree and pin multiplexing configuration.
- Debugged combined hardware/firmware issues with logic analyzer, oscilloscope and JTAG/SWD.
- Wrote Python scripts for automated firmware testing over serial connections.

Embedded Software Developer · IDF technological unit 2006 - 2012

- Developed real-time embedded software in C for communication equipment.
- Maintained an existing code base and performed debugging in field conditions.
- Completed the unit's embedded software training course; extended service track.

TECHNICAL SKILLS

MCU	ARM Cortex-M0/M3/M4/M7 (STM32, NXP)
RTOS	FreeRTOS, bare metal
Protocols	I2C, SPI, UART, CAN, BLE
Debug	JTAG/SWD, logic analyzer, oscilloscope
Tools	GCC ARM, CMake, Ceedling, Git, Jira
Other	Python (test scripts), C++ (basic)
Regulation	IEC 62304, work under ISO 13485 QMS

EDUCATION

B.Sc. Electrical Engineering

Ben-Gurion University, Beer Sheva

2012 - 2016

MILITARY SERVICE

IDF technological unit

Embedded software developer

2006 - 2012 (extended track)

LANGUAGES

Hebrew - native

English - good